

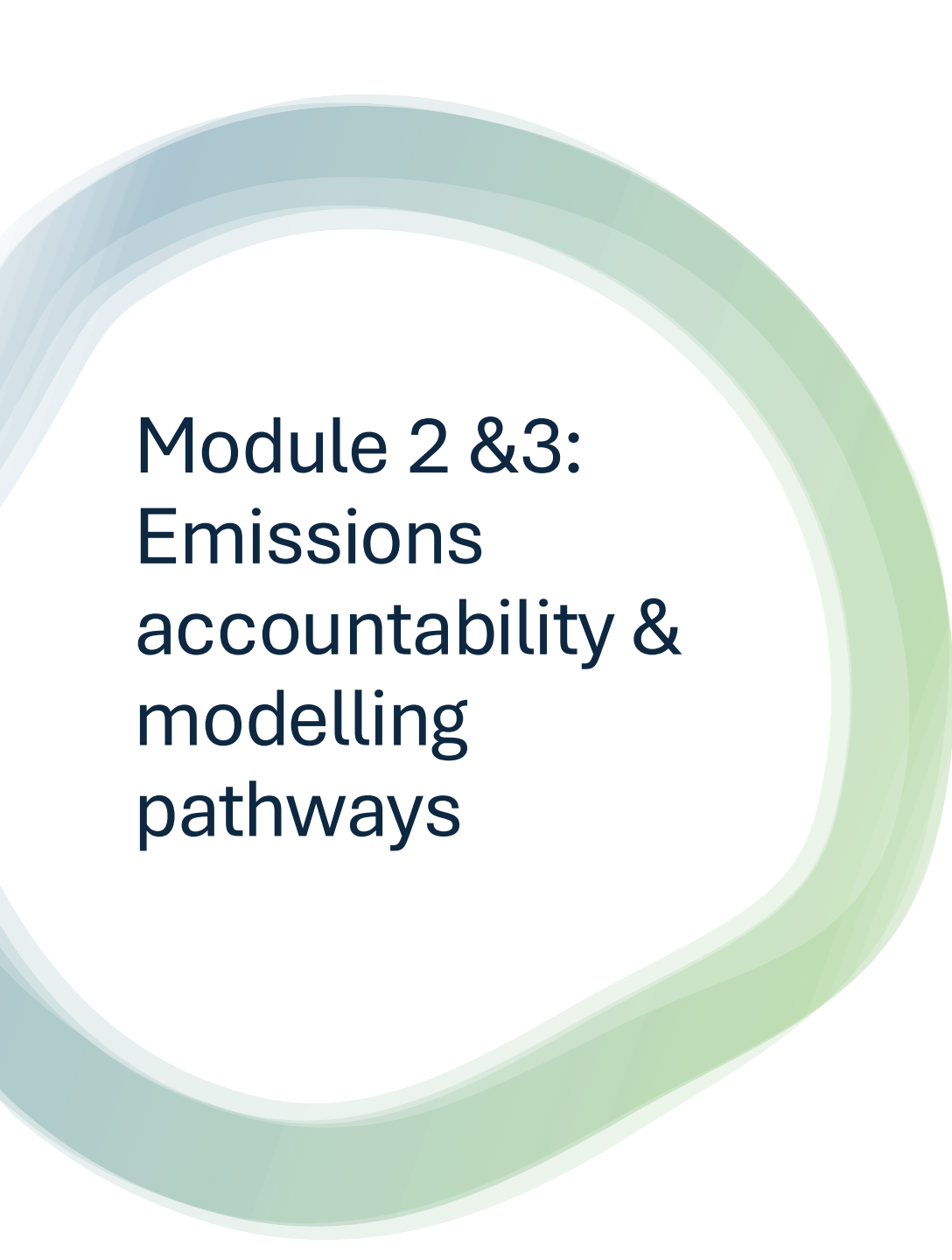
Climate Change Governance: Science, Data & Models



Lecture 6: IPCC & Scenarios

ENGG562/ENV562 – Fall 2026

Muhammad Awais



Module 2 &3: Emissions accountability & modelling pathways

- **Science & Negotiations** – CMIP, IPCC and other negotiation processes
- **Scenarios in Climate Change** – SSPs, RCPs and their role in assessing the future
- **Understand Systems Linkages** between the Anthropocene and the modern economy (Energy, Water, AFOLU)
- **Deep Dive into Integrated Assessment Models (IAMs)** – linking climate, energy, water, land, and development
- **Understand Policy Frameworks** and processes for Pakistan (e.g., NDCs, ETFs, accounting & reporting)
- **Design your Own Context-Specific scenarios** based on regional challenges and priorities

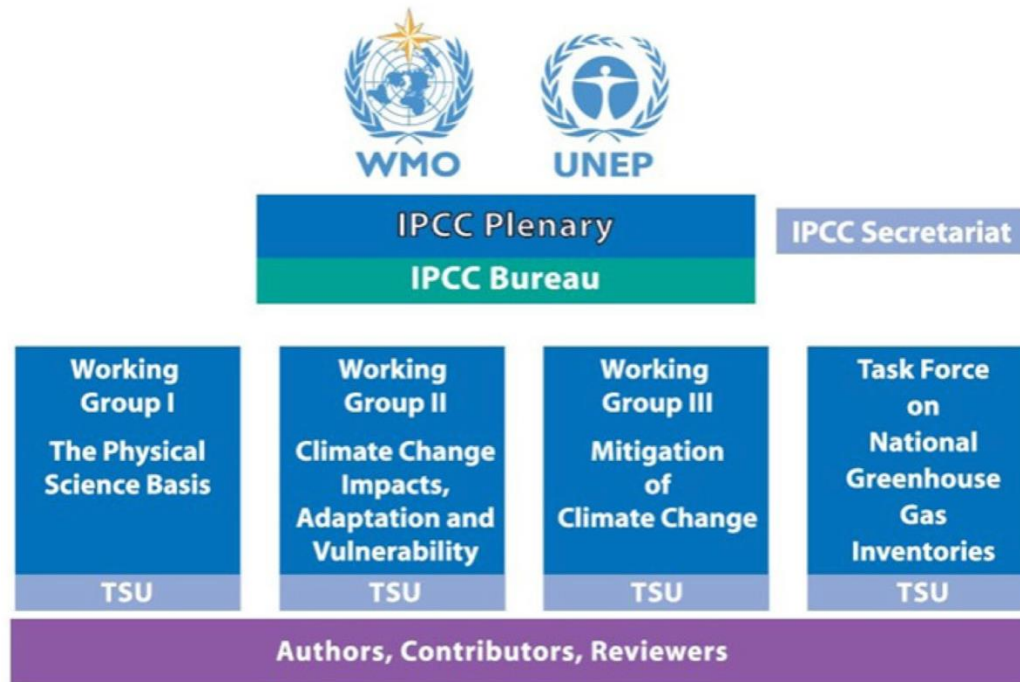
Introduction to the IPCC



- Leading international body for the assessment of climate change
 - 195 member countries
 - Created in 1988 by the WMO and UNEP
 - Task:
 - "to assess on a comprehensive, objective, open and transparent basis the scientific, technical and socio-economic information relevant to understanding the scientific basis of risk of human-induced climate change, its potential impacts and options for adaptation and mitigation.*
 - IPCC reports should be neutral with respect to policy, although they may need to deal objectively with scientific, technical and socio-economic factors relevant to the application of particular policies."*
- (see principles governing the IPCC's work)*

- IPCC is the primary source for scientific input into the UNFCCC
- Assessment of current understanding of climate change
 - Does not conduct new research
 - Reviews and assesses latest scientific information
- Advantage over other literature: reports are government approved
- Approval of summary report line by line by governments

Working modality of IPCC



- IPCC plenary comprises of all countries in the world
- IPCC Bureau comprises of 30 elected members with variable numbers for each of the WMO region; IPCC elects its bureau members once in a 6-7 years cycle
- 3 working groups & a Task Force on NGGI
- Authors, Contributors, Reviewers, Review Editors



Scoping

The outline is drafted and developed by experts nominated by governments and observer organizations



Approval of Outline

The Panel then approves the outline



Nomination of authors

Governments and observer organizations nominate experts as authors



Government and Expert Review - 2nd Order Draft

The 2nd draft of the report and 1st draft of the Summary for Policymakers (SPM) is reviewed by governments and experts



Expert Review - 1st Order Draft

Authors prepare a 1st draft which is reviewed by experts



Selection of authors

Bureaux select authors



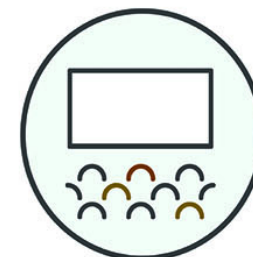
Final draft report and SPM

Authors prepare final drafts of the report and SPM which are sent to governments



Government review of final draft SPM

Governments review the final draft SPM in preparation for its approval



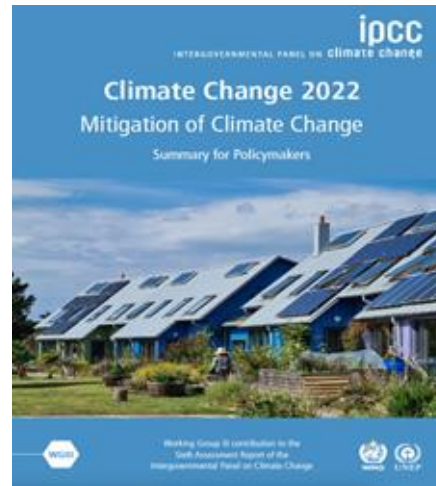
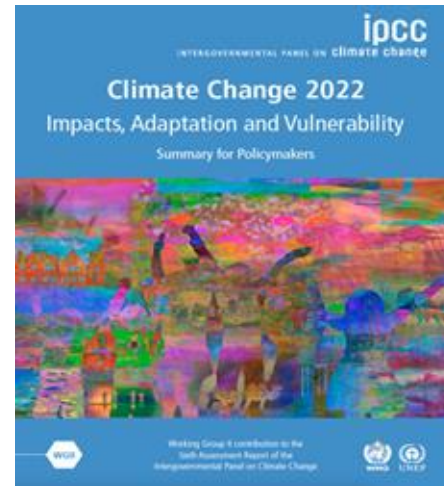
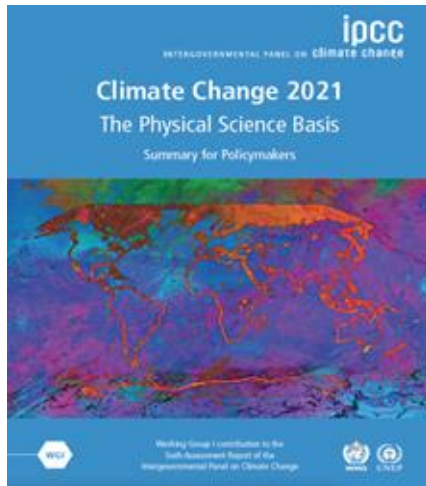
Approval & acceptance of report

Working Group/Panel approves SPMs and accepts reports

IPCC Publications

1. Assessment Reports:

- 3 volumes, together with their Summaries for Policymakers (SPM), plus a Synthesis Report (SYR)
 - WG1: physical science
 - WG2: impacts, adaptation and vulnerability
 - WG3: mitigation



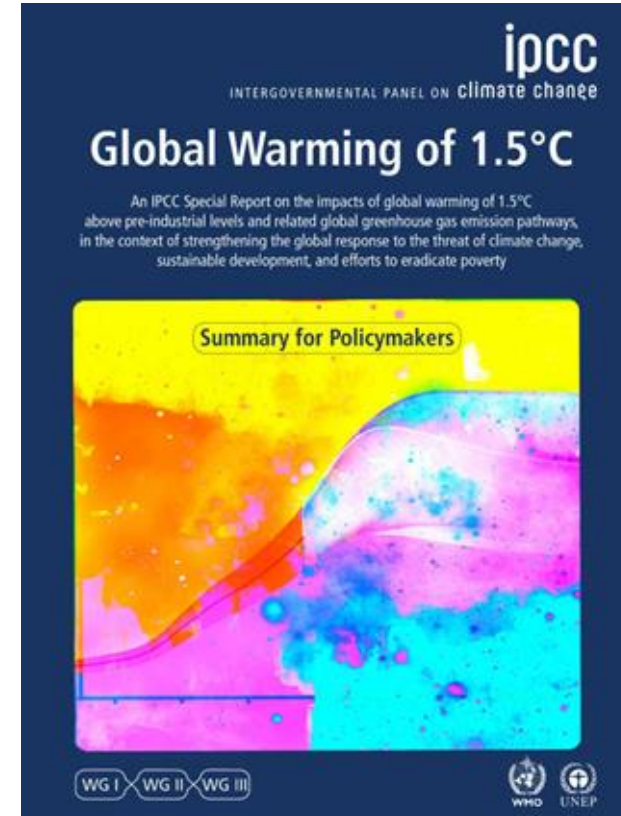
IPCC Publications

2. Special Reports:

- provide assessment of a specific issue
- Collaborations between WGs

3. Methodology Reports:

- practical guidelines for the preparation of greenhouse gas inventories



IPCC Report Authorship

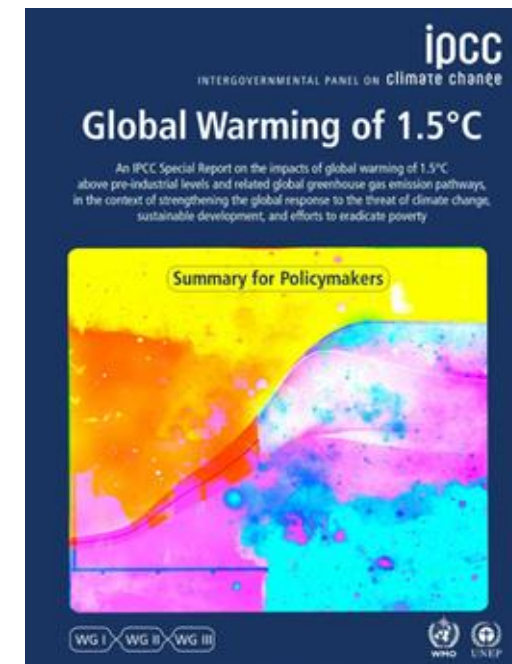
IPCC Report Authorship

- **Coordinating Lead Author (CLA)**
 - ~ 2 CLA/chapter, developing/developed country
 - coordinate content
- **Lead Author (LA)**
 - ~10-20 LA/chapter
 - produce content of the chapter
- **Review Editor (RE)**
 - ~2 RE/chapter
 - ensure review comments are addressed by the author teams, ensure diversity in literature is reflected adequately
- **Contributing Author**
 - Experts asked to contribute to specific parts of chapters
- **Chapter Scientist**
 - ~1/chapter
 - Assist authors, literature reviews, organize information

The Special Report on 1.5°C shows the risks of exceeding 1.5°C (in particular compared to 2°C) and confirms feasibility of limiting warming to 1.5°C.

Core findings to inform the UNFCCC:

- **Where we are now:** how much has the globe warmed so far, and what are the impacts?
- **Where do we want to be:** what is the gap between current climate change action and what is needed to limit warming to 1.5°C?
- **And how we could get there:** what options are available for ambitious climate action?



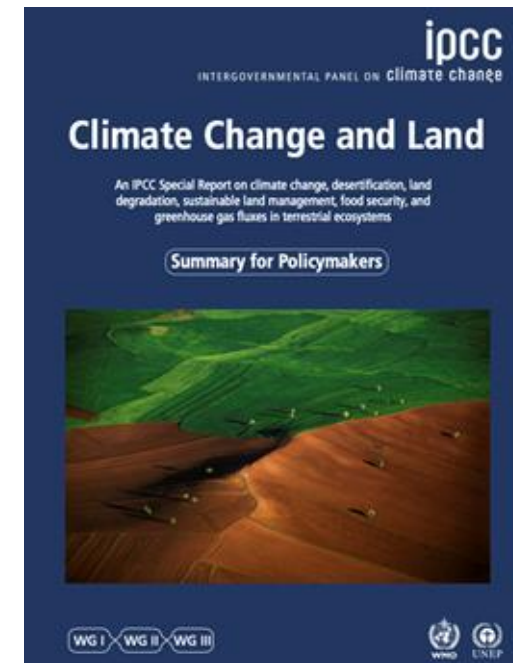
2018

Highlights from the IPCC AR6 cycle

The Special Report on Climate Change and Land (SRCCL) highlights linkages between climate, land degradation and food security.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- **Decarbonisation needed across all sectors**, only land-based mitigation insufficient to achieve 1.5°C
- **More nuanced assessment of response options**
- Increased **focus on 1.5°C** specific information
- **Regional heterogeneity of risks**



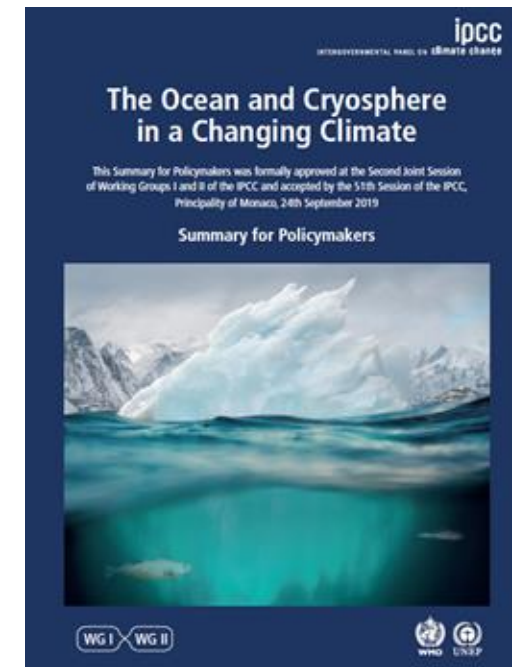
2019

Highlights from the IPCC AR6 cycle

The Special Report on Cryosphere and Oceans (SROCC) emphasises the long climate legacy of our actions for oceans and ice sheets.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- Inclusion of **multi-century sea level rise** projections
- **Planned relocation and forced migration** not adaptation options but responses
- **Regional information relevant to developing countries** strengthened and safeguarded



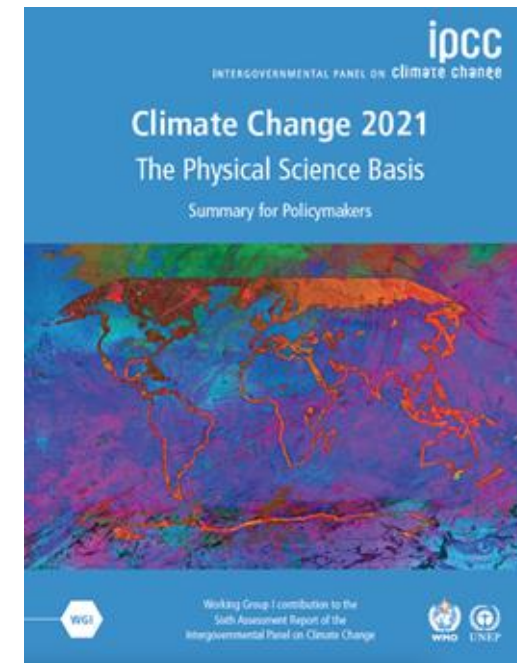
2019

Highlights from the IPCC AR6 cycle

The WGI report confirms the grave risks facing us in a warming world.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- **methodological changes** in the **historical temperature assessment**
- **1.5°C crossing times**
- relevant **regional information** and long-term **sea level rise** assessment

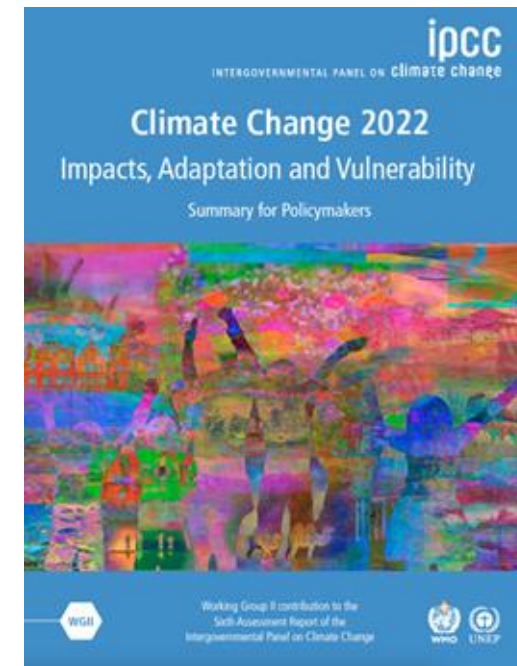


Highlights from the IPCC AR6 cycle

The WGII report shows that loss and damage is already a reality for vulnerable countries.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- **”losses and damages”**
- scenario information with a focus on **1.5°C**
- Transformative adaptation
- solar radiation modification with a critical lens

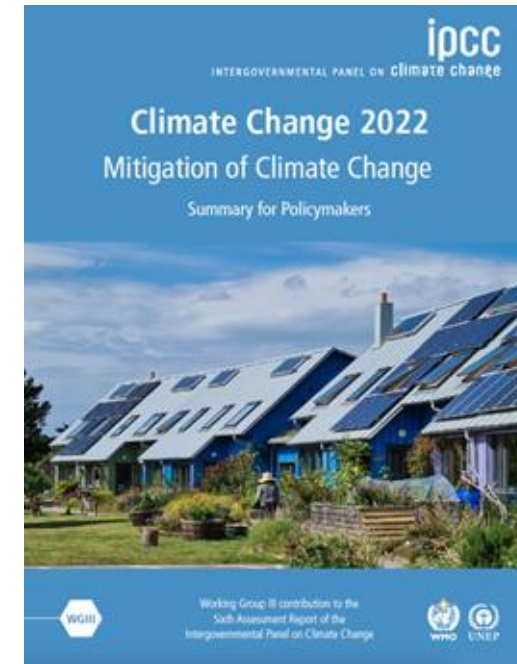


Highlights from the IPCC AR6 cycle

The WGIII report shows that it is still possible to limit warming to 1.5°C, and what is needed to get there.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- **1.5°C**, with the creation of a new pathway category on "1.5°C with no or limited overshoot and with net zero GHGs" to reflect the PA
- mitigation options and **renewable energy**, and necessary reductions in **fossil fuels**



2022

Highlights from the IPCC AR6 cycle

The Synthesis Report (SYR) confirms that there is no justification for delay in this critical decade.

Delegates contributed to improvements in the representation of core findings on: *(selection)*

- rephrasing the statements on **reaching 1.5°C "in the near term"** for careful messaging
- detailed information on **emissions reductions to limit warming to 1.5°C, including for 2035**
- expanded information on **"losses and damages", adaptation limits, sea level rise**
- statements specific to SIDS and LDCs



2023

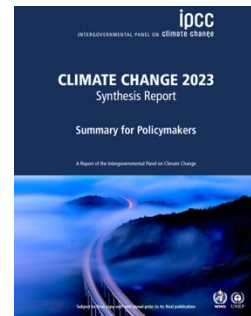
Milestone reports establishing our best available scientific knowledge on climate change



2001 “There is new and stronger evidence that most of the warming observed over the last 50 years is attributable to human activities”

1995 “The balance of evidence suggests a discernible human influence on global climate”

1990 “... humankind is capable of raising the global-average annual-mean surface-air temperature...”



2023 “Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming, with global surface temperature reaching 1.1° C above 1850-1900 in 2011-2020.”

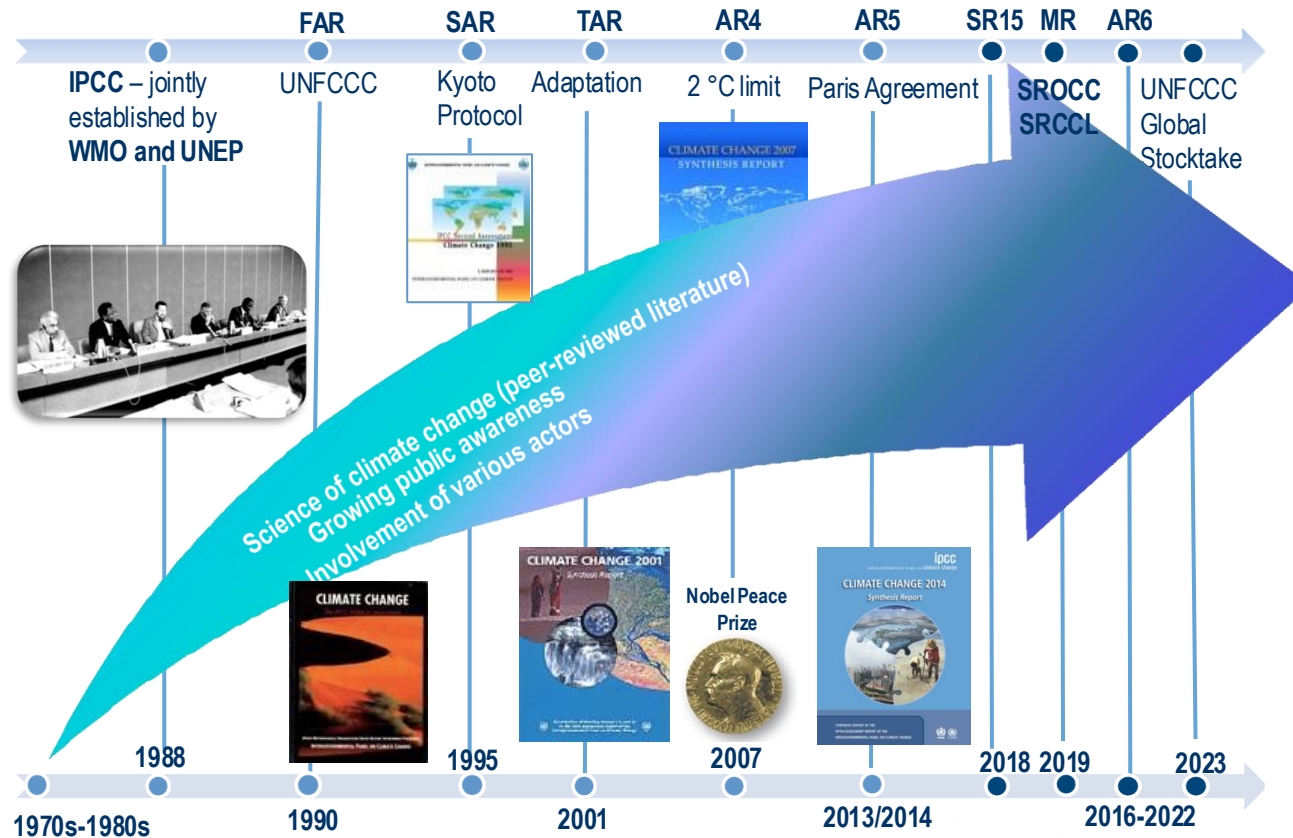


2018 “Human activities are estimated to have caused approximately 1.0 degree Celsius of global warming above pre-industrial levels, ...”

2013 “Human influence on the climate system is clear.”

2007 “Warming of the climate system is unequivocal...”

IPCC: Assessment reports



What is the IPCC working on at the moment?

The 7th assessment cycle:

ipcc
REPORTS SYNTHESIS REPORT WORKING GROUPS ACTIVITIES
NEWS CALENDAR

NEWSROOM POST
IPCC invites government representatives and experts to review the second draft of the Special Report on Climate Change and Cities

ipcc

REPORTS SYNTHESIS REPORT WORKING GROUPS ACTIVITIES NEWS
CALENDAR

Seventh Assessment Report

The IPCC is currently in its seventh assessment cycle which formally began in July 2023 with elections of the new Chair and new IPCC and TFI Bureaus.

AGREED OUTLINES

AUTHORS

CarbonBrief

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Ongoing failure to agree AR7 timeline is 'unprecedented' in IPCC history

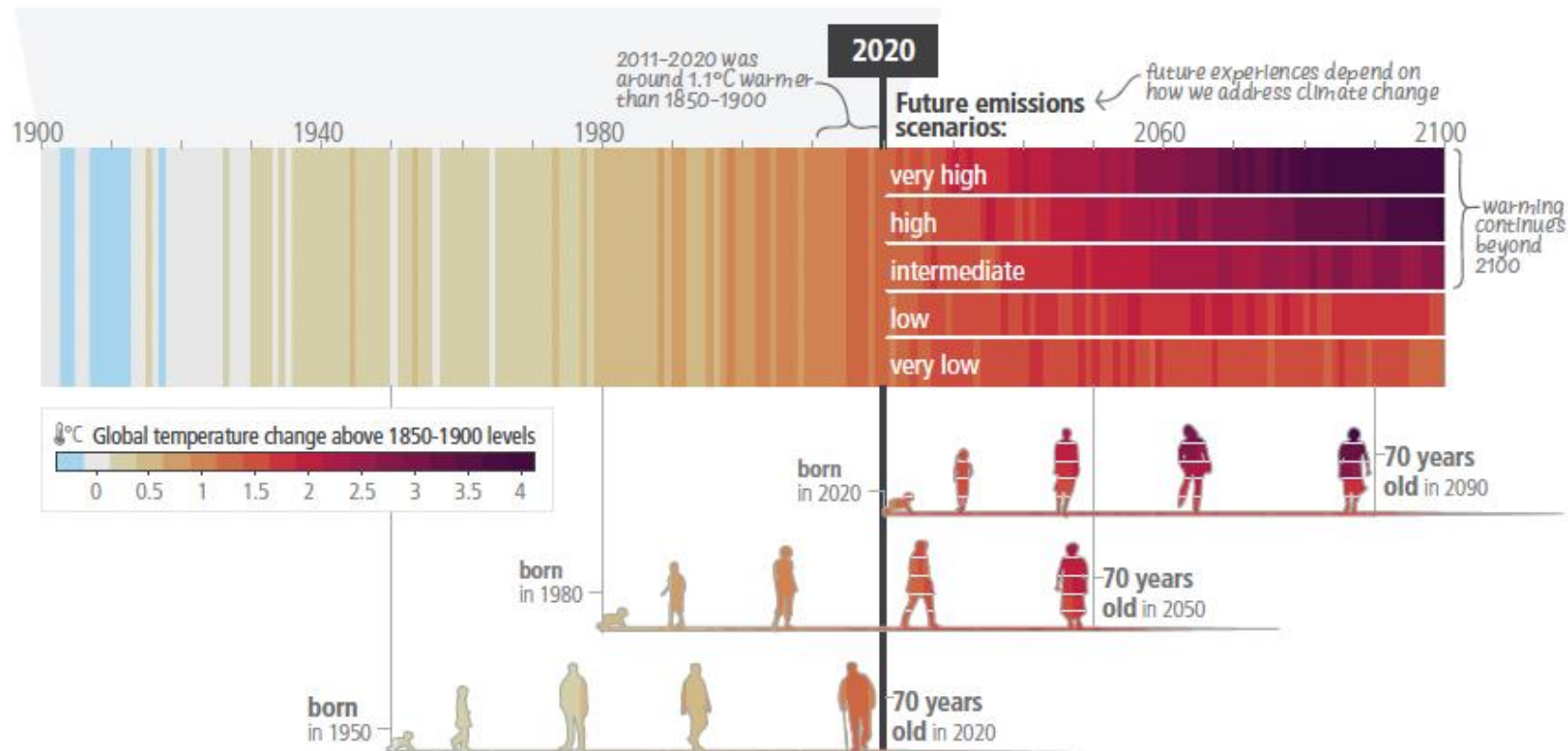
07.11.25 | AR7

By: Aneha Tandon, Cecilia Keating, Robert McSwanney

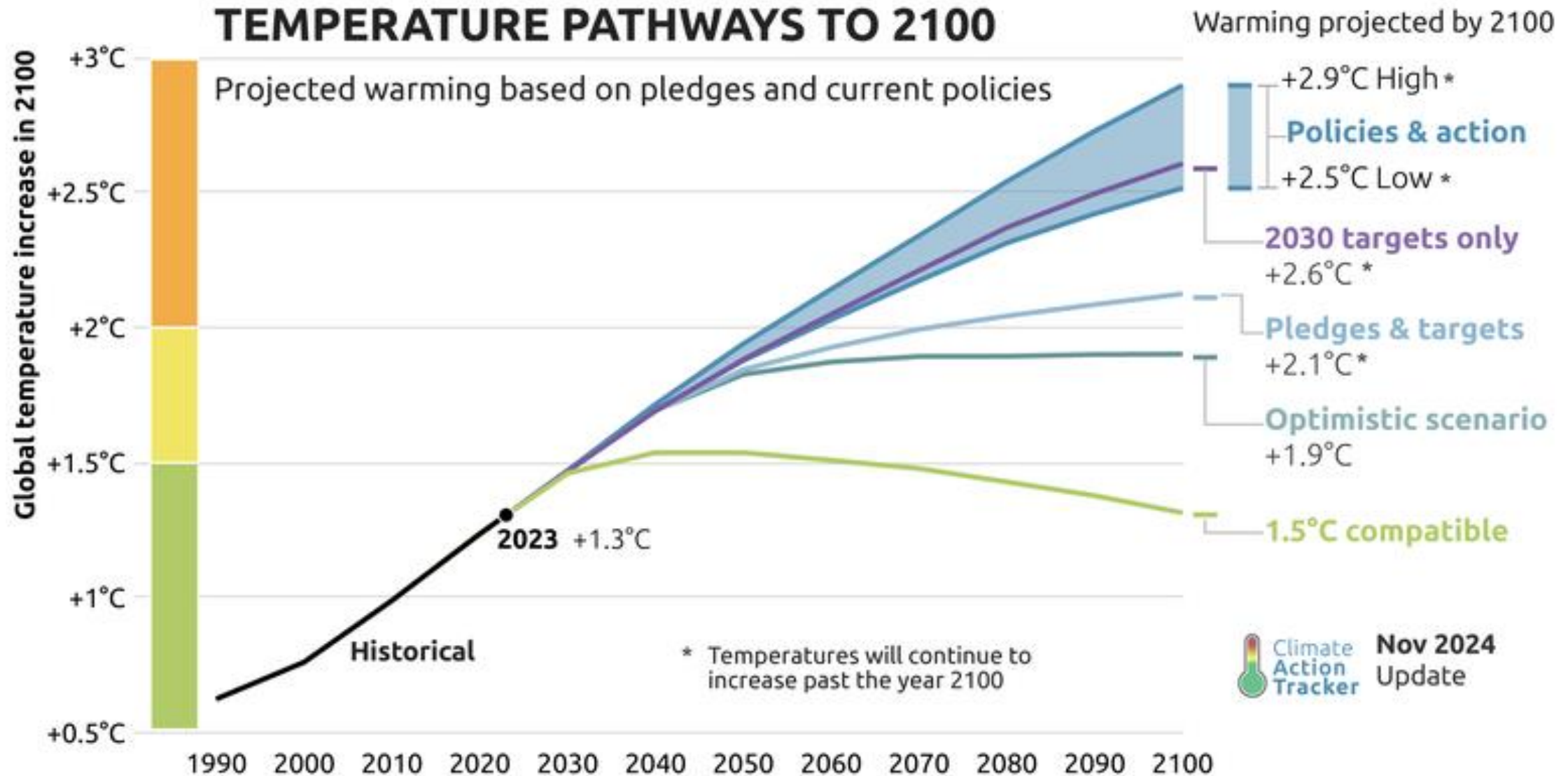


Observed and projected warming – where does it come from ?

The extent to which current and future generations will experience a hotter and different world depends on choices now and in the near term

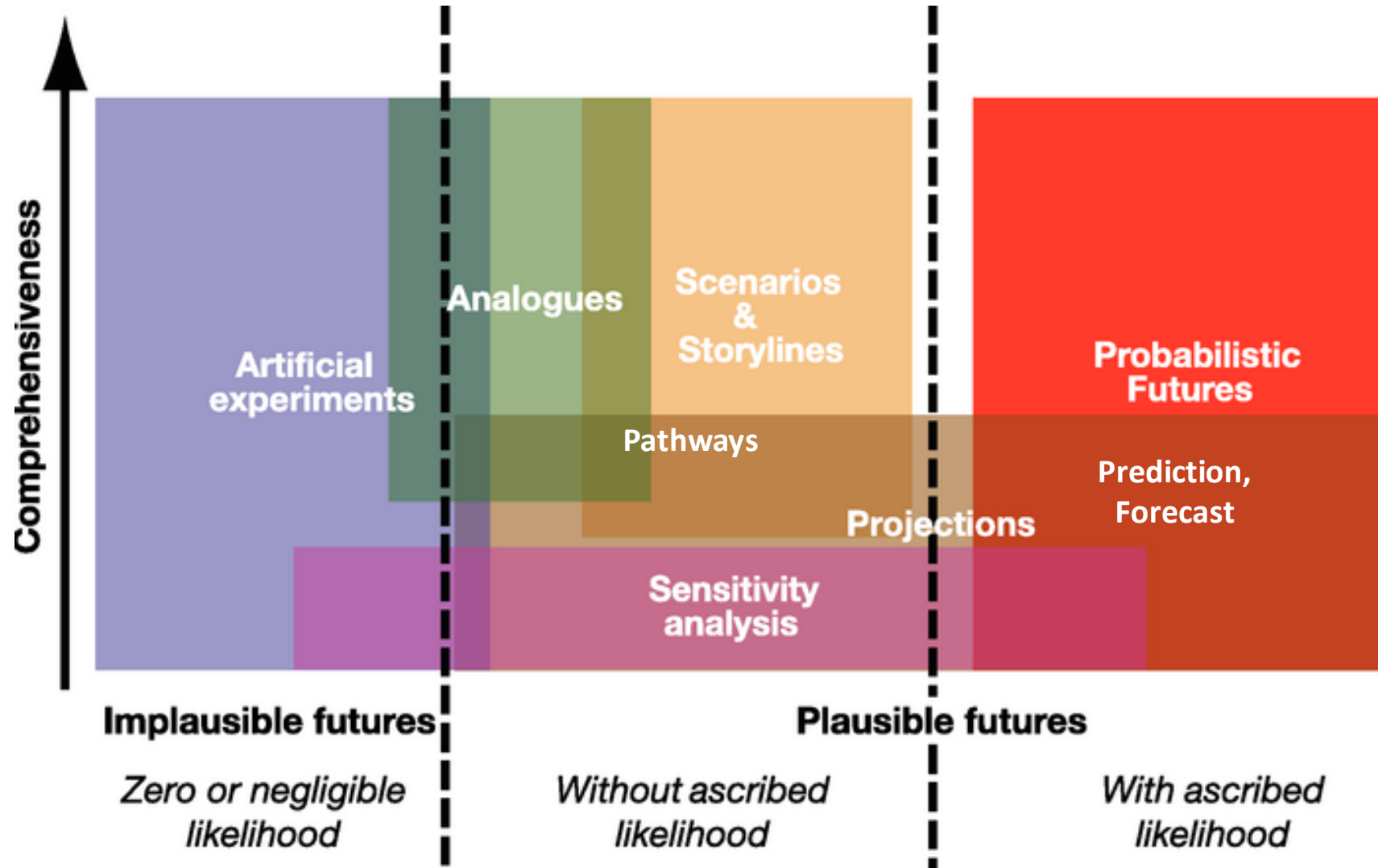


Where do these trajectories come from ?



Characterizations of the future

Any structured description of how the world might look in coming decades



What are scenarios, why do we use them?

Scenarios.

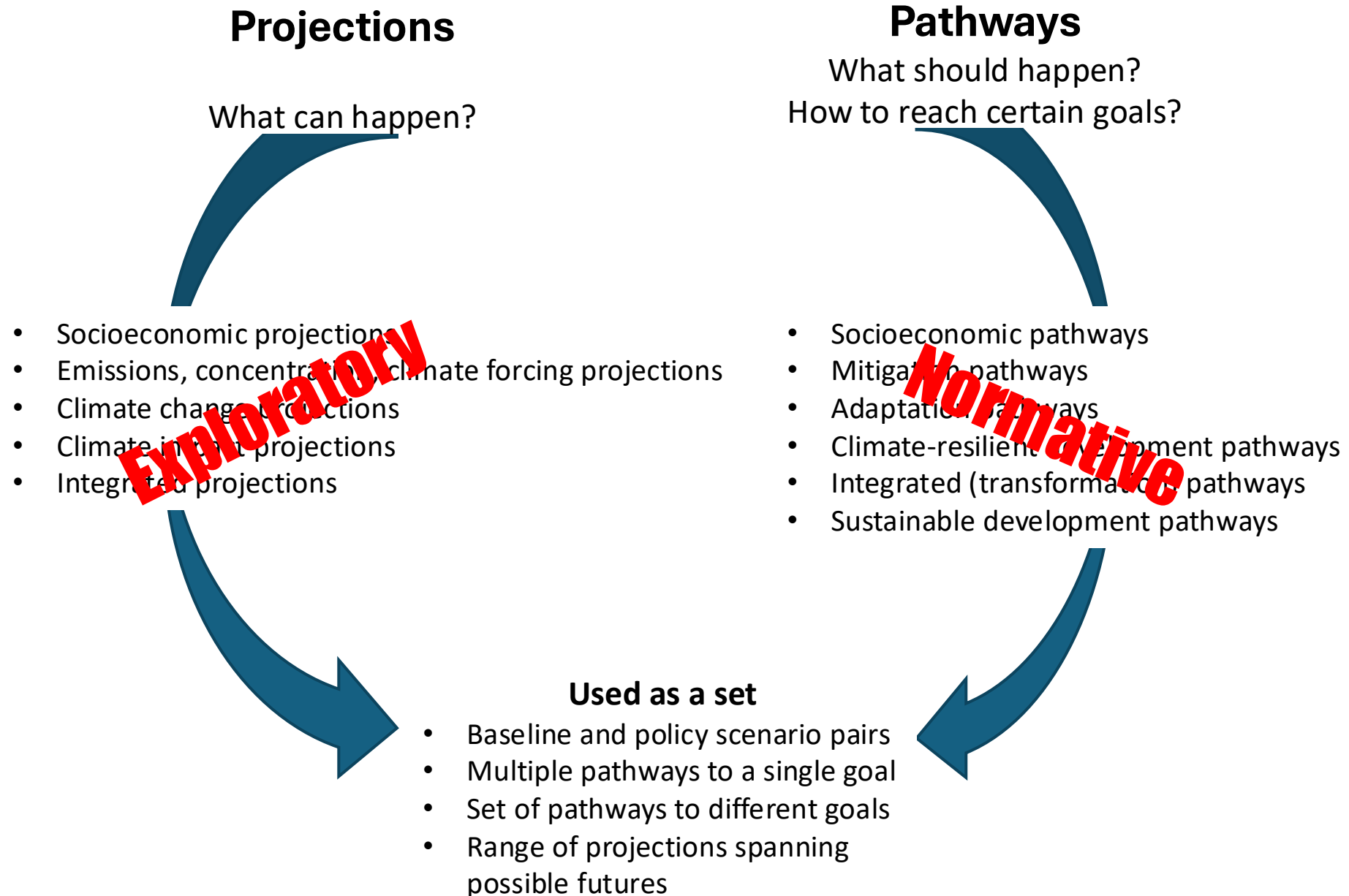
A scenario is a coherent, internally consistent, and **plausible** description of a possible future state of the world.

Scenarios are **not predictions or forecasts** (which indicate outcomes considered most likely) but are alternative images without ascribed likelihoods of how the future might unfold.

They may be **qualitative, quantitative, or both**.

An **overarching logic** often relates several components of a scenario, for example a storyline and/or projections of elements of a system.

IPCC: Climate Change Scenarios



Source: Elmar Kriegler, SENSES project

Scenarios are what-if storylines about future

- They describe different possible futures under certain assumptions (e.g., population, economy, technology, or policies).
- They are not predictions, but rather a way to explore uncertainty of future.
- **Examples:**
 - *Everyday:* Imagine a scenario where **50%** of students at LUMS commute by public transport instead of private vehicles.
 - *In context of Climate Change:* Imagine a scenario where Pakistan decides that **50%** of all new motorbike sales must be electric.

Projections are model results that show what could happen *if a given scenario comes true*

- They are the numerical outcomes, such as changes in temperature, rainfall, or energy use.
- **Examples:**
 - *A simple example would be* If you save **PKR100** each day, the projection is that you will have **PKR3,000** at the end of the month.
 - In context of climate change, If average summer temperature in Lahore is **40°C**, a projection under a high fossil-fuel use scenario might show it rising to about **44–45°C** by the end of the century.

Models and Scenarios

- Models are structured relationships that produce **conditional forecasts**.
- Scenarios are **conditional** forecasts.

